AAA-B-105B
December 31, 1970
SUPERSEDING
Fed. Spec. AAA-B-105A
January 25, 1966 and
Fed. Spec. AAA-B-96C
July 15, 1959

FEDERAL SPECIFICATION

BALANCES, PRESCRIPTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers prescription balances with capacities of 120 grams and 4 Troy ounces.

1.2 Classification.

1.2.1 Types and classes. Prescription balances covered by this specification shall be of the following types and classes, as specified (see 6.2).

Type I - 120 gram capacity, permanently assembled.

Type II - 4 Troy ounce capacity, knock down.

Class 1 - 15 milligram sensitivity.

Class 2 - 30 milligram sensitivity.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

AAA-W-200	-	Weight Sets, Balance
PPP-B-1122	-	Balances, Scales and Accessories, Packaging and Packing Of

FSC 6670

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Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H44 - Specification, Tolerances and Other Technical Requirements for Commercial Weighing and Measuring Devices.

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3. REQUIREMENTS

3.1 Materials. Materials shall be of the highest quality throughout as normally used by the manufacturer, provided the end item complies with all the provisions of the specification.

3.2 Design and construction. The type I and type II balances shall meet the applicable specifications specified in NBS H44 for prescription balances, unless otherwise specified herein. The type I balance shall have a nominal capacity of 120 grams. The type II balance shall have a capacity of 4 Troy ounces. The sensitivity requirement of the type I balance shall conform to the sensitivity requirement (SR) for Class A prescription balances specified in NBS H44 (see 4.3.1). The type II, class 1 balance shall have a sensitivity requirement of 15 milligrams (mg), and the type II, class 2 balance shall have a sensitivity requirement of 30 mg (see 4.3.1).

3.2.1 Construction, type I. The type I balance shall be permanently assembled, mounted in a case, and have a fully enclosed lever system with stabilized pans mounted above the beam. Adjustable balancing means and a balance indicator shall be provided. A mechanical means of arresting and releasing the lever system, manually operated leveling feet, and a level indicator shall be provided. The balancing and adjusting screws shall be brass or other nonmagnetic metal, finished to resist corrosion. Weighing shall be accomplished by counterbalance with weights of known value, by reading directly from a weighbeam scale, by reading from a graduated dial, or by a combination of these methods.

3.2.1.1 Balance case and cover. The balance case shall contain the lever system and shall be constructed of glass and wood. Alternatively, the case may be constructed of glass and metal with a corrosion-resistant finish. A case cover of the same materials as the case shall be hinged to and, when closed, shall rest evenly on the case top. The cover shall have a transparent glass top. The case cover shall have a glass front, when required for viewing the indicators and scales.

3.2.1.2 Weigh pans. Weigh pans shall be polished, corrosion-resisting steel, or chromium plated brass. The pans shall be concave and circular in shape with an intergral projection to form a handle. Reversal of the pans on the pan holders shall not affect the condition beyond the point that the balance cannot be brought to the equilibrium position with the zero adjustment mechanism, when tested as specified in 4.3.2.

3.2.1.3 Adjusting weights. Poise adjusting weights shall be of the one piece construction and shall be fabricated of corrosion-resisting steel or metal with a corrosion-resistant finish.

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3.2.1.4 Poise indicator. A poise indicator shall be provided to indicate the equilibrium of the balance. The indicator shall consist of a fixed scale and single or double pointers.

3.2.1.5 Arresting and releasing mechanism. A manually operated mechanism for arresting and releasing the lever system shall be provided. The control for operating the mechanism shall be mounted on the outside of the balance case. The arresting and releasing mechanism, when tested as specified in 4.3.3, shall not affect the equilibrium condition of the balance.

3.2.1.6 Weigh beams or graduated dials. Weigh beams or graduated dials shall have both an Apothecaries and a Metric scale. The Apothecaries scale shall be subdivided into graduations of not more than 1/8 grain and shall have a capacity of not less than 10 grains. The metric scale shall be subdivided into graduations of not more than 10 mg and shall have a capacity of not less than 650 mg. Weigh beam riders shall be of corrosion-resisting steel, or of metal treated to resist corrosion, and shall be positioned on the beam by mechanical means, manually operated from outside the balance case.

3.2.2 Construction, type II. The type II balance shall be an equal arm scale, and shall be mounted in a case. The balance shall be capable of being disassembled and the disassembly placed within the drawer of the case. The balance shall have a column and a crossbeam with a pan suspended from each end with chains. There shall be a balance indicator and an arrest mechanism. A weight set conforming to type V of AAA-W-200 shall be furnished with each balance. The case shall be made of wood and the drawer either of wood or aluminum. The pans and chains shall be constructed of brass. The balance shall be constructed to allow disasembly in five parts; (1) the case, including the column mounting flange and drawer; (2) the column, including the arrest mechanism and graduated scale; (3) the crossbeam, including the pointer; (4) left pan and chains; and (5) right pan and chains.

3.2.2.1 Case. The case shall have a flange attached to the top to secure the column in the mounted position and shall have a drawer opening to the front. The drawer shall have blocking to secure the weight set and the disassembled balance parts in position. Means shall be provided to secure the drawer in the closed position. The case of the class 1 scale shall be a minimum of 12 inches long, by 8-1/2 inches wide, by 4-1/2 inches high. The case of the class 2 scale shall be a maximum of 9-1/16 inches long, by 5-1/2 inches wide, by 3-3/8 inches high.

3.2.2.2 Column. A tubular column shall attach to the mounting flange. The column shall contain a shaft which shall move vertically in the column when actuated by the lifting device. The beam fulcrum bearing shall be attached to the shaft. Means shall be provided to prevent rotation of the shaft.

3.2.2.3 Crossbeam. The crossbeam shall be a minimum of 6 inches in length, centered on the column, and shall pivot on a knife edge and a bearing.

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3.2.2.4 Pans and chains. The pans shall be a minimum 3-1/4 inches in diameter, and shall be a minimum of 3/4 inch deep. There shall be three chains of equal length secured equidistant on each pan circumference. The other end of the chains shall be joined together and suspended from the crossbeam.

3.2.2.5 Arrest mechanism. The arrest mechanism shall consist of a vertical shaft in the tubular column and a lifting mechanism operated from outside the column to raise and lower the shaft. When the shaft is raised, the pans shall not touch the top of the case, and when the shaft is lowered, the pans shall rest on the top of the case. The shaft shall have a bearing attached for the crossbeam.

3.2.2.6 Balance indicator. The balance indicator shall consist of a pointer extending from the beam and a graduated scale at the top of the column. The graduated scale shall have a minimum of four graduations on each side of a zero graduation. The division between graduations shall be a minimum of 0.08 inch. The graduations shall be uniform in size and character.

3.2.2.7 Bearings and knife edges. The knife edges shall be steel and the bearings shall be steel or agate. Bearings shall be secured in position.

3.2.2.8 Overload. The balance, with a load of 8 Troy ounces in each pan, shall show no bends, deformations or misalignment of bearings or knife edges when tested as specified in 4.3.4.

3.2.2.9 Tolerance. The weighing tolerance shall be not more than 0.1 percent of the test load, when tested as specified in 4.3.5.

3.3 Finish. The case, if wood, shall be stained and varnished. All brass parts, including weights, shall be lacquered or plated. Other metal parts, except aluminum weights, pivots, and bearings of the beam, shall be rendered corrosion-resistant by suitable coatings.

3.4 Marking for identification. Each balance shall be marked with a capacity and the manufacturer's name or trademark, model number, and serial number. In addition, type I balances shall be marked "class A".

3.5 Repair and maintenance. Provisions shall be made to allow for adjustment, service, and replacement of components. In addition, there shall be free access to inspect, service, and adjust any component as provided for by this specification or as indicated by the component or equipment manufacturer, without requiring disassembly of any other component on the unit.

3.6 Service data. Service data shall be supplied, as specified (see 6.2).

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3.7 Workmanship. The finished balance shall have no burrs, slivers, or splinters. The drawer shall open and close without binding. The balance shall be rigid, when assembled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and materials inspection. In accordance with 4.1 above, components and materials shall be tested and inspected in accordance with all the requirements of referenced specifications and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. A lot shall consist of all balances of one type and class, as applicable, offered for inspection at one time. The sample unit for inspection shall be one completely fabricated balance.

4.2.3 Visual examination. Examination of the end item shall be in accordance with the classification of defects specified in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished as specified.		X
Construction and workmanship (applicable to all components and assemblies)	Component missing, improperly assembled, fractured, or malformed.	X	
	Not free of burrs, slivers, or splinters.		X

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TABLE I. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Type I			
Weight beam or dial scales	Not combination (Metric and Apothecaries) type, as specified.	X	
	Scales not graduated as specified.	X	
	Weigh beam poise or dial not operable from outside balance case.	X	
Balance indicator	Not as specified.	X	
Case cover	Does not rest evenly on case top when closed.	X	
	Does not permit clear view of level indicator, balance indicator, and weigh beam scales when in a closed position.	X	
Arresting means	Not operable from outside of case.	X	
Leveling feet	Not provided.	X	
Level indicator	Not provided.	X	
Type II			
Case	Drawer does not open or close.	X	
	Drawer cannot be secured.		X
	Blocking not as specified.		X
Column	Not secured to the flange; i.e., rotates.	X	
Crossbeam	Does not pivot.	X	
	Not centered.	X	
Pans and chains	Chains not equidistant on pan.	X	
	Chains not equal in length.	X	
Arrest mechanism	Not operated from outside the column.	X	
	Does not raise or lower the pans.	X	

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TABLE I. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Type II (cont'd)			
Balance indicator	Less than four graduations on each side of zero graduation.	X	
	Graduations not uniform in size and character.		X
Bearings and knife edges	Bearings not secured.	X	
	Bearings not steel or agate.	X	
	Knife edges not steel.	X	
Marking identification (type I and type II)	Missing, incomplete, not legible.		X
Service data (when specified)	Missing, incomplete, not legible.	X	

4.2.4 Dimensional examination, type II. Inspection shall be made of the type II end item for compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.2.5 Testing of the end item. The end item shall be tested for the characteristics specified in 4.3.1 through 4.3.5, as applicable. The inspection level shall be S-2. Failure of one or more sample units to meet all test requirements shall be cause for rejection of the lot.

4.2.6 Examination of packaging and packing. Examination of packaging and packing shall be made in accordance with PPP-B-1122.

4.3 Tests.

4.3.1 Sensitivity. The sensitivity of the type I balance shall be tested with loads of 30, 60, 90, and 120 grams. Weights equal to 1/2 these loads shall be placed in each weigh pan to bring the balance indicator to zero. A weight equal to the sensitivity requirement, specified in NBS H44 for class A prescription balances, shall be placed in either one of the weigh pans. The type II balances shall be tested for sensitivity requirement at zero load and at loads of 1, 2, and 3 Troy

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ounces, and capacity load. Weights equal to 1/2 these loads shall be placed in each pan. A 1/4 grain weight for the class 1, or a 1/2 grain weight for the class 2 balance as applicable, shall be applied to one pan at each of the test loads. Failure of the balance indicator to indicate an imbalance condition shall constitute failure of this test.

4.3.2 Weigh pans. The weigh pans shall be tested by placing the pans on the pan holders and noting the position of the balance indicator. The pans shall be reversed on the pan holders and the position of the balance indicator observed to determine compliance with 3.2.1.2. Any nonconformance shall constitute failure of this test.

4.3.3 Arresting means, type I balance. The arresting and releasing means for the type I balance shall be tested under loads of 30, 60, 90, and 120 grams. Weights equal to 1/2 these loads shall be placed in each weigh pan and the position of the balance indicator noted. The loads shall be arrested and released 10 times each. The balance indicator shall be observed before arresting and after releasing to determine compliance with 3.2.1.5. Any nonconformance shall constitute failure of this test.

4.3.4 Overload, type II balance. The overload test shall be performed before any other end item test is performed. The arrest mechanism shall be released and an 8 Troy ounce load applied to each load receiver and then removed. The balance shall be examined before and after removing the weight to determine compliance with the requirements specified in 3.2.2.8. Any nonconformance shall constitute failure of this test.

4.3.5 Tolerance, type II balance. The type II balance shall be tested for tolerance at loads of 25, 50, and 124 grams. Weights equal to 1/2 of these loads shall be applied to each pan and the weight required to obtain the equilibrium position shall be observed. Any nonconformance to the requirements of 3.2.2.9 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing for level A, B, or C shall be in accordance with PPP-B-1122 (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The balances covered by this specification are intended for use in weighing the ingredients for medical prescriptions in any amount up to 120 grams. Type II balances are intended for general purpose weighing in dental laboratories and field locations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents, as applicable (see 6.1).

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2.1).
(c) Service data required (see 3.6).
(d) Levels of packaging and packing required (see 5.1).
(e) When special marking is required for civil agencies (see 5.2.1).

6.3 Classification comparison. The balances covered by AAA-B-105A and AAA-B-98C are included in this specification as follows:

AAA-B-105A	Type I
AAA-B-98C, class 1	Type II, class 1
AAA-B-98C, class 2	Type II, class 2

MILITARY CUSTODIANS:

Army - GL
Navy - MS
Air Force - 03

Review activities:

Army - MD
Defense Supply Agency - DM

User activity:

Army - SM

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Balances, Prescription AAA-B-105B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

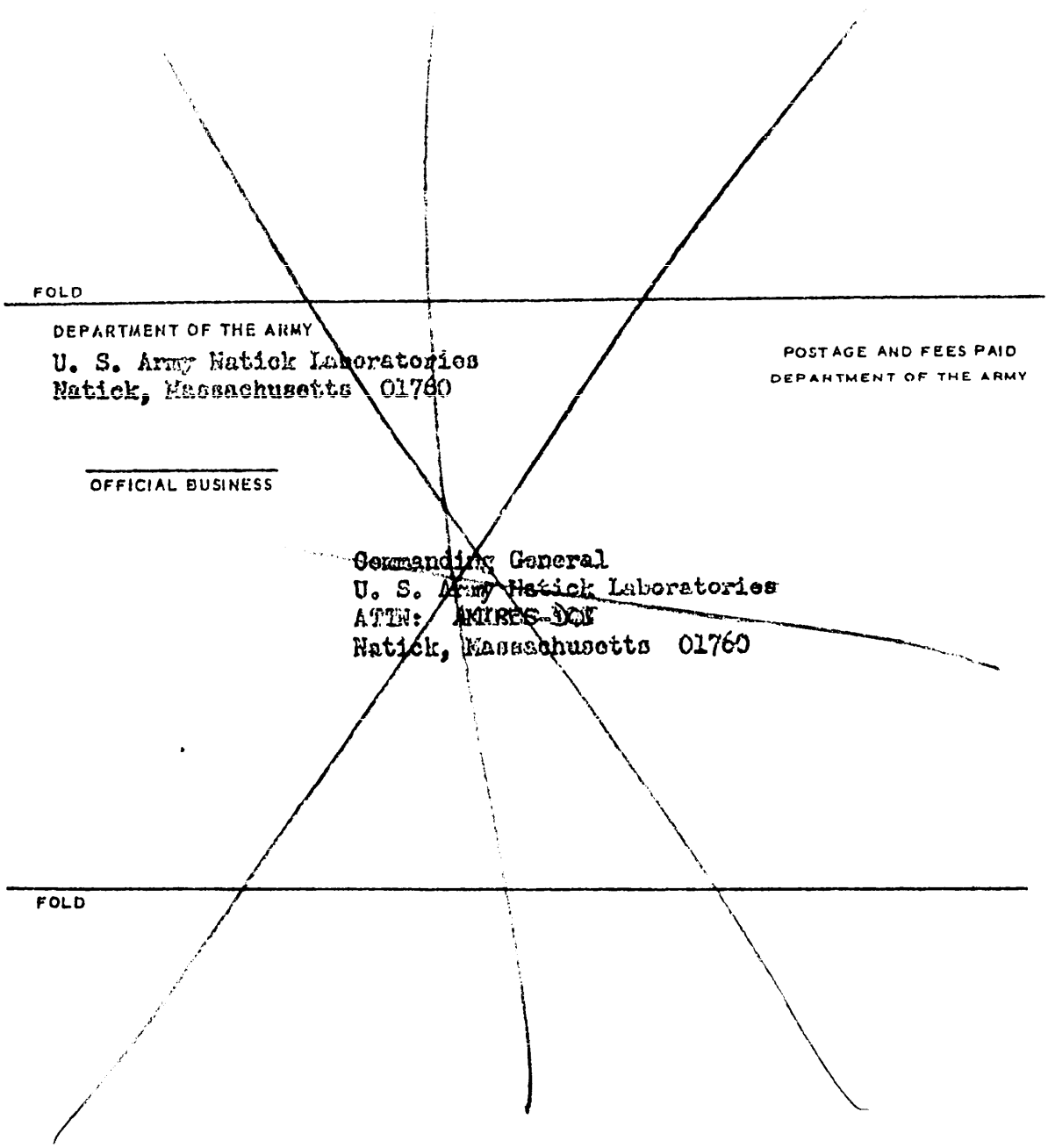
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